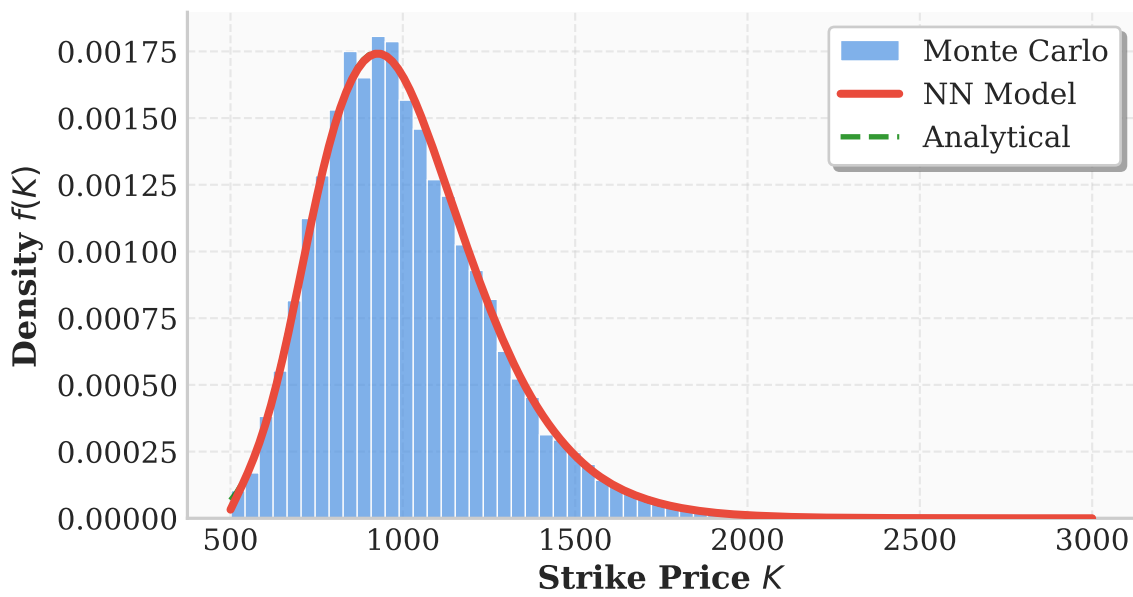


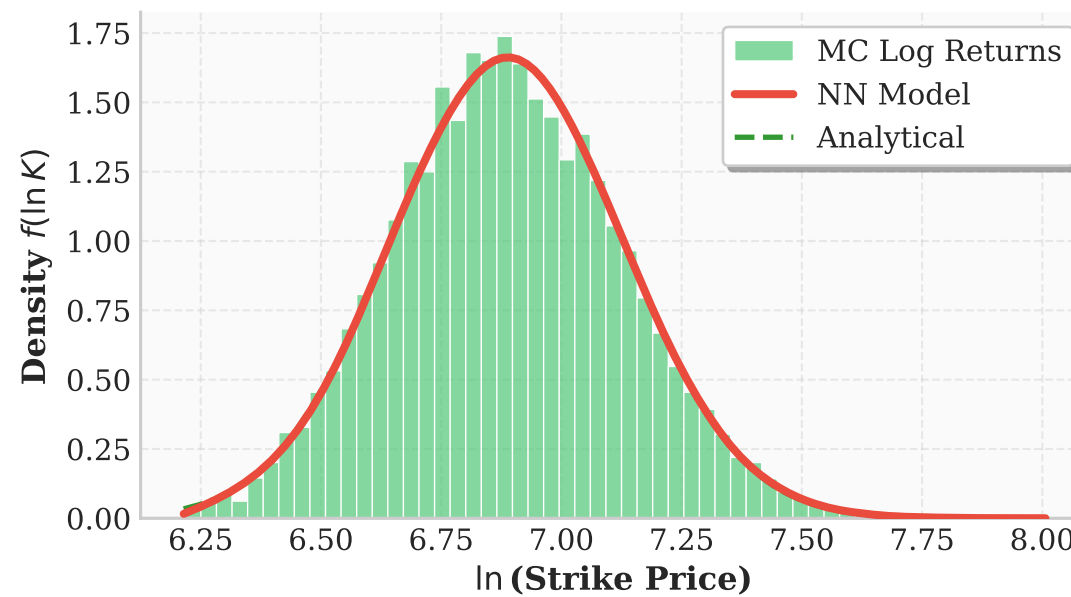
PDF Analysis: Neural Network Synthetic Local Volatility Model

Monte Carlo with $dS_t = rS_t dt + \sigma_{NN}(t, S)S_t dW_t$

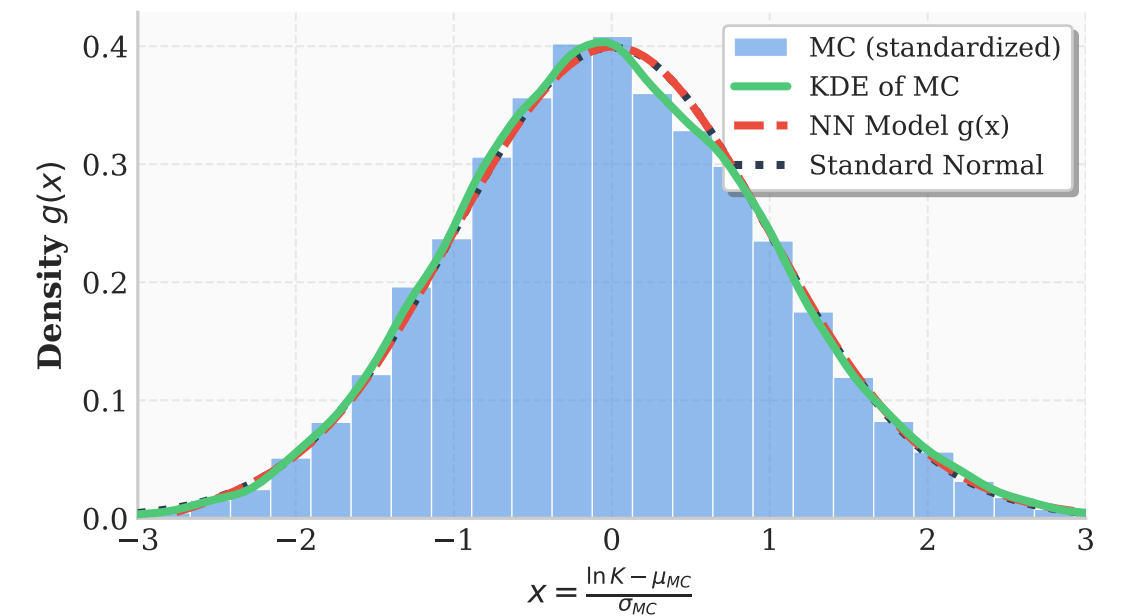
Strike Distribution (T = 0.25)



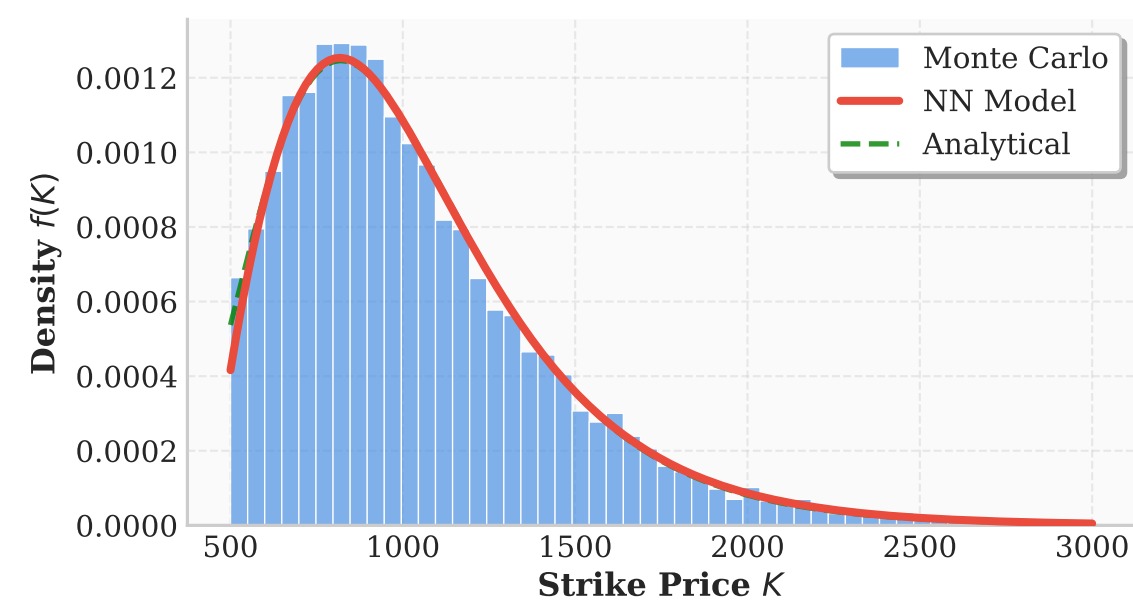
Log-Normal Distribution (T = 0.25)



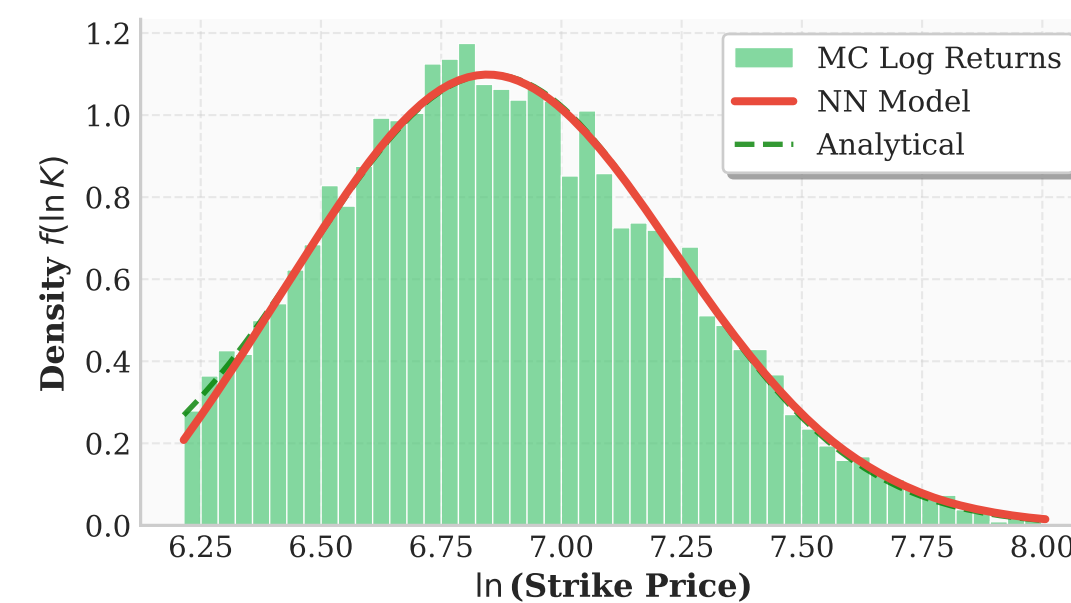
Gaussian Space (T = 0.25)
MC Standardization: $\mu_{MC}=6.89, \sigma_{MC}=0.24 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.89, \sigma_{model}=0.24$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=0.08, \text{ExKurt}=-0.01$



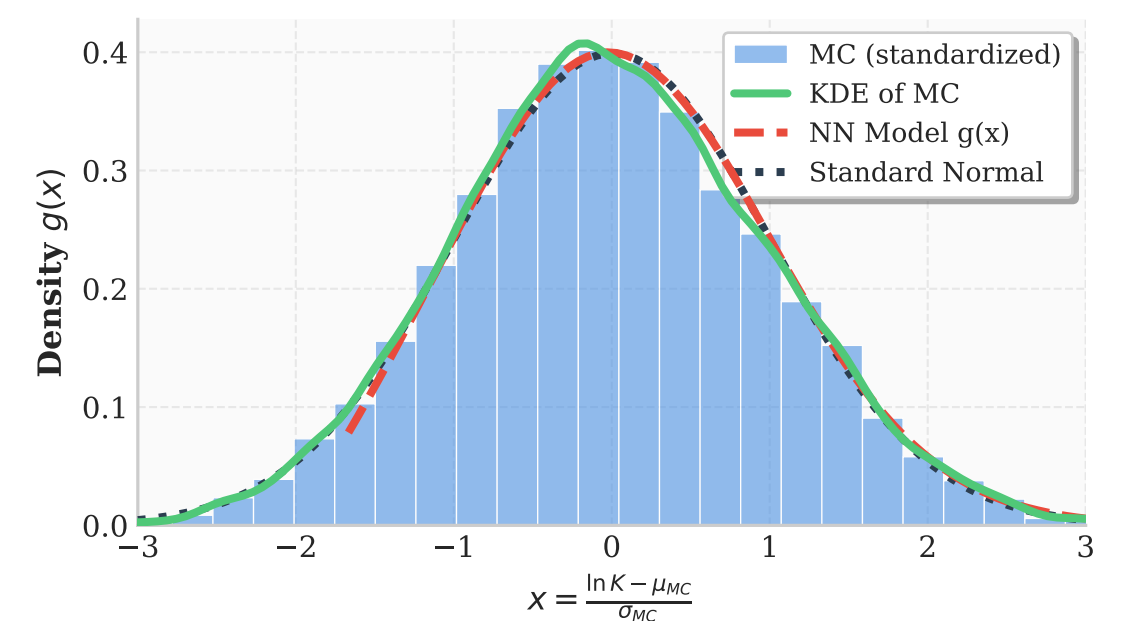
Strike Distribution (T = 0.50)



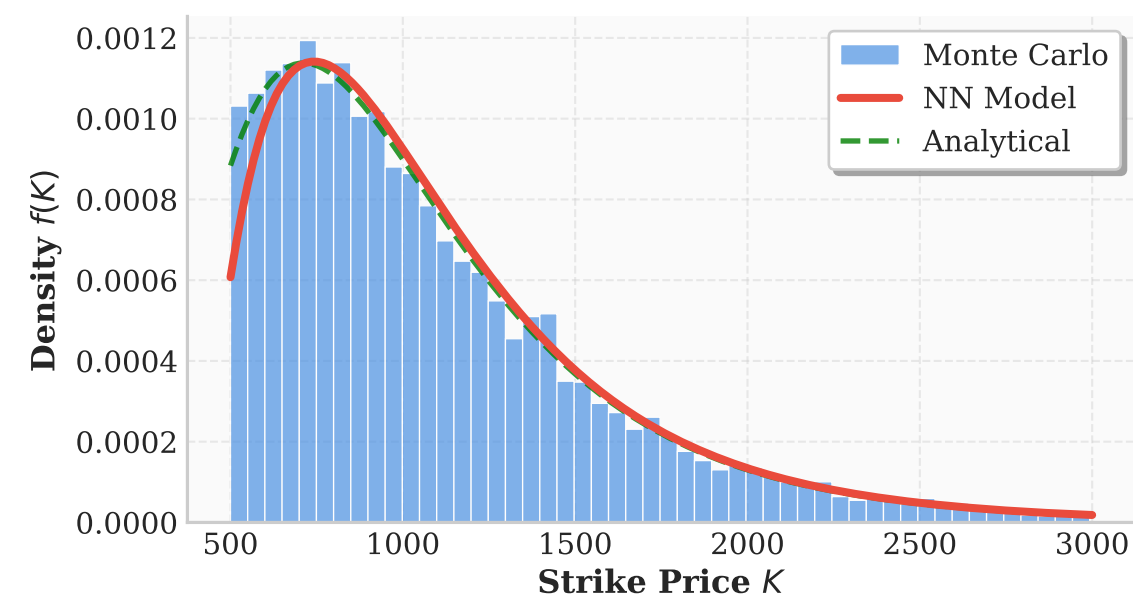
Log-Normal Distribution (T = 0.50)



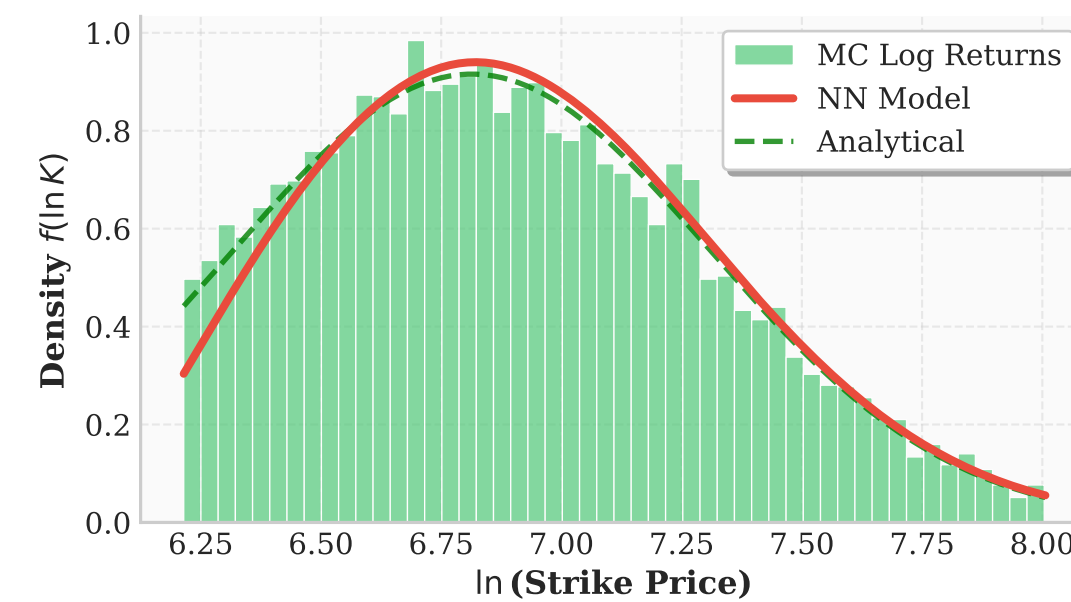
Gaussian Space (T = 0.50)
MC Standardization: $\mu_{MC}=6.85, \sigma_{MC}=0.38 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.90, \sigma_{model}=0.36$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=0.09, \text{ExKurt}=-0.01$



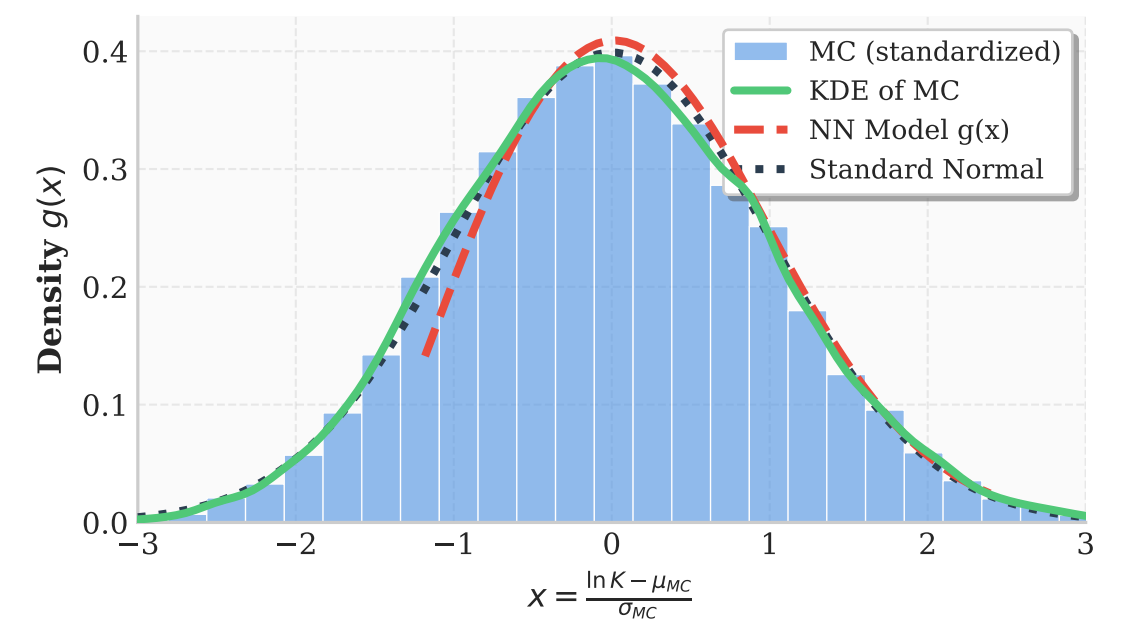
Strike Distribution (T = 0.75)



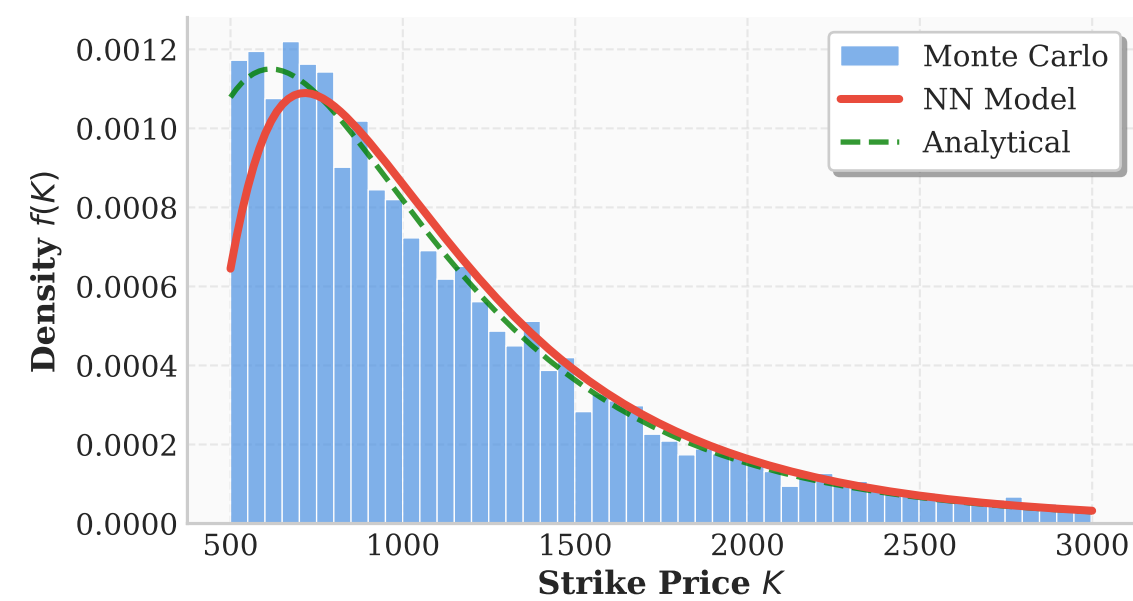
Log-Normal Distribution (T = 0.75)



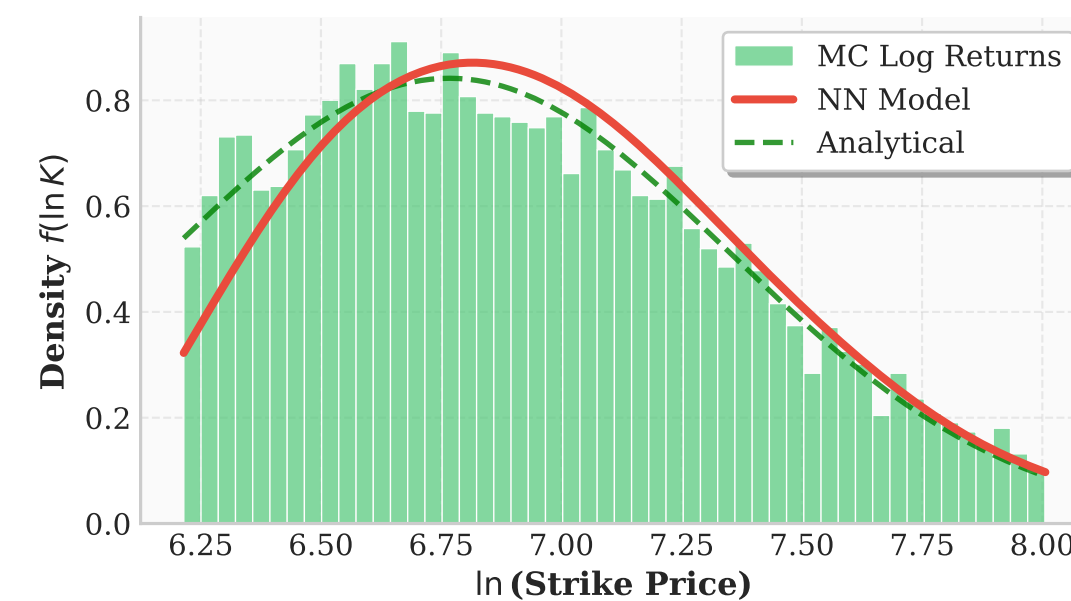
Gaussian Space (T = 0.75)
MC Standardization: $\mu_{MC}=6.81, \sigma_{MC}=0.50 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.92, \sigma_{model}=0.41$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=0.12, \text{ExKurt}=-0.07$



Strike Distribution (T = 1.00)



Log-Normal Distribution (T = 1.00)



Gaussian Space (T = 1.00)
MC Standardization: $\mu_{MC}=6.77, \sigma_{MC}=0.59 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.95, \sigma_{model}=0.43$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=0.10, \text{ExKurt}=-0.08$

